MILITARY SPECIFICATION

PRIMER ELECTRIC, M83
METAL PARTS FOR (ALTERNATE)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for an alternate design of Primer, Electric, M83.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provision, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARMY

9327339 - Head Primer
9327340 - Body
9327342 - Plug, Closing
8839509 - Retainer
8831022 - Plug

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts components.-The metal parts components shall comply with all requirements specified on applicable drawings and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.5.1 (see 6.4).

3.4 Workmanship.-All parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection.-The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer. The first article sample shall consist of ten (10) complete sets of parts (i.e., ten (10) each of every component). The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with lot provisions of MIL-A-48078.

4.4.2 Examination.-Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63306 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Metal Parts Components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Head Primer (dwg. 9327339) Examination for Defects	10		3.2	4.4.2.1
	Retainer (dwg. 8839509) Examination for Defects	10		3.2	4.4.2.2
	Closing Plug (dwg. 9327342) Examination for Defects	10		3.2	4.4.2.3
	Plug (dwg. 8831022) Examination for Defects	10		3.2	4.4.2.4
	Body (dwg. 9327340) Examination for Defects	10		3.2	4.4.2.5

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.1	Head Primer		9327339
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of external threads, min.		0.40%	3.2	Gage
102	Major diameter of external threads, min.		0.40%	3.2	Gage
103	Pitch diameter of large internal threads, max.		0.40%	3.2	Gage
104	Minor diameter of large internal threads, max.		0.40%	3.2	Gage
105	True position of pitch diameter of large internal threads with pitch diameter of external threads		0.40%	3.2	Gage
106	Pitch diameter of small internal threads, max.		0.40%	3.2	Gage
107	Minor diameter of small internal threads, max.		0.40%	3.2	Gage
108	True position of pitch diameter of small internal threads with diameter of large unthreaded cavity		0.40%	3.2	Gage
109	Flange diameter		0.40%	3.2	Gage
110	True position of flange diameter with pitch diameter of external threads		0.40%	3.2	Gage
111	Flange thickness		0.40%	3.2	Gage
112	Diameter of large unthreaded cavity		0.40%	3.2	Gage
113	Diameter of small unthreaded cavity		0.40%	3.2	Gage
114	True position of diameter of small unthreaded cavity with diameter of large unthreaded cavity		0.40%	3.2	Gage
115	Protective coating missing		0.40%	3.2	Visual
116	Protective coating inadequate		See 4.5.2	3.2	4.5.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET	2 OF 3	9327339
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pitch diameter of external threads, max.		0.65%	3.2	Gage
202	Pitch diameter of large internal threads, min.		0.65%	3.2	Gage
203	Pitch diameter of small internal threads, min.		0.65%	3.2	Gage
204	Length, total		0.65%	3.2	Gage
205	Length from flange to beginning of external threads		0.65%	3.2	Gage
206	Length from flange to end of external threads		0.65%	3.2	Gage
207	Depth of small unthreaded cavity		0.65%	3.2	Gage
208	Diameter, forward end, of large threaded cavity		0.65%	3.2	Gage
209	Outside diameter, forward		0.65%	3.2	Gage
210	Depth of large threaded cavity		0.65%	3.2	Gage
211	Minimum effective length of large internal thread		0.65%	3.2	Gage
212	Depth from front face to rear of small threaded cavity		0.65%	3.2	Gage
213	Countersink angle at the rear of small threaded cavity		0.65%	3.2	Gage
214	Width of wrench slots (if applicable)		0.65%	3.2	Visual
215	Depth of wrench slots (if applicable)		0.65%	3.2	Gage
216	Distance between wrench slots (if applicable)		0.65%	3.2	Gage
217	Depth of wrench holes (if applicable)		0.65%	3.2	Gage
218	Diameter of wrench holes (if applicable)		0.65%	3.2	Gage
219	Distance between wrench holes (if applicable)		0.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63306 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9327339 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
220	Distance from center hole to wrench hole(s) (if applicable)		0.65%	3.2	Gage
221	Chamfers missing or incorrect		0.65%	3.2	Visual
222	Radii missing or incorrect		0.65%	3.2	Visual
223	Surface finish incorrect		0.65%	3.2	Visual
224	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Retainer	1 OF 1	8839509
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, min.		0.40%	3.2	Gage
102	Major diameter, min.		0.40%	3.2	Gage
103	Diameter of hole, min.		0.40%	3.2	Gage
Minor					
201	Thickness, min.		0.65%	3.2	Gage
202	Slot missing		0.65%	3.2	Visual
203	Surface finish improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.4	Visual
205	Pitch diameter, max.		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63306 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plug, Closing				9327342
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, min.		0.40%	3.2	Gage
102	Hole missing or obstructed		0.40%	3.2	Gage
103	Major diameter, min.		0.40%	3.2	Gage
104	Thread length of engagement		0.40%	3.2	Gage
Minor					
201	Pitch diameter, max.		0.65%	3.2	Gage
202	Diameter of hole, min.		0.65%	3.2	Gage
203	Diameter of counterbore		0.65%	3.2	Gage
204	Total length, max.		0.65%	3.2	Gage
205	Length to shoulder, min.		0.65%	3.2	Gage
206	Length of thread relief		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63306 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Plug		8831022
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, min.		0.40%	3.2	Gage
102	Major diameter, min.		0.40%	3.2	Gage
103	Protective finish missing		0.40%	3.2	Visual
104	Protective finish inadequate	See 4.5.2		3.2	4.5.2
Minor					
201	Thickness		0.65%	3.2	Gage
202	Width of slot		0.65%	3.2	Gage
203	Depth of slot		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.4	Visual
205	Pitch diameter, max.		0.65%	3.2	Gage

NOTES:

10

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Body				9327340
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Flash hole(s) missing		100%	3.2	Gage
2	Hydrostatic (Prior to machining)		100%	3.3	4.5.1
3	Cracks		100%	3.2	Test (Non-destructive) See Note 1
Major					
101	Plug gage fails to pass through body		0.40%	3.2	Gage
102	Location of first row of holes		0.40%	3.2	Gage
103	Location of second row of holes		0.40%	3.2	Gage
104	Wall thickness		0.40%	3.2	Gage
105	Diameter of flash holes		0.40%	3.2	Gage
106	Pitch diameter of external thread, min.		0.40%	3.2	Gage
107	Major diameter of external thread, min.		0.40%	3.2	Gage
108	Pitch diameter of internal thread, max.		0.40%	3.2	Gage
109	Minor diameter of internal thread, max.		0.40%	3.2	Gage
110	True position of flash holes		0.40%	3.2	Gage
111	Protective finish missing		0.40%	3.2	Visual
112	Protective finish inadequate	See 4.5.2		3.2	4.5.2
113	Outside diameter		0.40%	3.2	Gage

NOTES:
1. Recommend either eddy current or magnetic particle.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63306 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET				9327340
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Total length		0.65%	3.2	Gage
202	Length of internal thread		0.65%	3.2	Gage
203	Length to end of external threads		0.65%	3.2	Gage
204	Length of thread pilot		0.65%	3.2	Gage
205	Diameter of thread pilot		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.4	Visual
207	Pitch diameter of external threads, max.		0.65%	3.2	Gage
208	Pitch diameter of internal threads, min.		0.65%	3.2	Gage

NOTES:

12

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures

4.5.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

$$P = \frac{1.8\ St}{D}$$

Where:

P = tests pressure in pounds per square inch (psi)
S = minimum specified yield strength, psi
t = specified minimum wall thickness in inches
D = specified maximum outside diameter in inches

4.5.2 Protective coatings.-Procedures for testing, the cleaning of components and the application of protective coatings shall be in accordance with the criteria for testing as specified on the applicable component drawings and in the specification referenced thereon.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The parts shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use in the M83 electric primer (alternate). This is used on 105MM Cartridges, M456A1 (HEAT) and M490 (TP).

6.2 Ordering data.-See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval - (See MIL-A-48078).-Submit equipment designs as required, to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Hydrostatic test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Custodian:
ARMY-AR

Preparing Activity:
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Project Number: 1390-A199

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63306 Primer Electric, M83 Metal Parts For (Alternate)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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